

Element G: Construction of a Testable Prototype

Construction Log:

- Day 1–3: Frame and float test on surface (no load)
- Day 4–6: Thruster mounts and wiring, dry test with Arduino
- Day 7: Submersion with tether, partial depth test
- Day 8: Failed deep test (flip event), logged behavior and tether load

Prototype was fully testable with realistic components and all systems mounted.

Functional: 6-motor drive, full control input response, modular float system. All parts (thrusters, frame, controller) are modeled on production-ready marine robotics kits.